

PNST EXAM PREPARATION

Organic Chemistry

Complete Notes for Weak Students

Simple Language • Key Reactions • Memory Tricks • Quick Revision

■ **MNEMONIC:** These notes are specially designed for students who find organic chemistry difficult. Each chapter uses simple language, highlights the most important reactions, and includes memory tricks (mnemonics) to help you remember concepts easily.

Table of Contents

Ch#	Topic	Pg
Ch 1	Basic Principles of Organic Chemistry (GOC)	2
Ch 2	Hydrocarbons	5
Ch 3	Haloalkanes & Haloarenes	9
Ch 4	Alcohols, Phenols & Ethers	12
Ch 5	Aldehydes, Ketones & Carboxylic Acids	15
Ch 6	Amines	19
Ch 7	Organic Compounds Containing Nitrogen	22

Chapter 1

Basic Principles of Organic Chemistry

General Organic Chemistry (GOC) & Reaction Mechanisms

1.1 What is Organic Chemistry?

Organic chemistry is the study of carbon-containing compounds. Carbon is special because it can form 4 bonds and long chains — this gives millions of compounds.

■ **FORMULA/RULE:** Carbon always makes EXACTLY 4 bonds. Hydrogen makes 1. Oxygen makes 2. Nitrogen makes 3.

1.2 Hybridization — Simple Rules

Hybrid.	Bond type	Shape	Angle	Example
sp^3	4 single bonds	Tetrahedral	109.5°	Alkanes (CH_4)
sp^2	1 double bond + 2 single	Trigonal Planar	120°	Alkenes, Benzene
sp	1 triple bond + 1 single	Linear	180°	Alkynes

■ **FORMULA/RULE:** More s-character $\rightarrow$ shorter bond, stronger bond, more acidic H.

1.3 Inductive Effect (I Effect)

Electron density shifts through sigma (σ) bonds due to electronegativity difference.

- **+I effect** (electron donating): Alkyl groups ($-CH_3 > -CH_2CH_3 > \dots$) push electrons $\rightarrow$ increase electron density.
- **-I effect** (electron withdrawing): Halogens, $-NO_2$, $-CN$ pull electrons $\rightarrow$ decrease electron density.

■ **FORMULA/RULE:** -I order of halogens: $-F > -Cl > -Br > -I$

1.4 Resonance Effect (Mesomeric Effect)

Electron delocalization through π bonds or lone pairs. The real structure is a hybrid of all resonance structures — it is NOT one of them.

- **+M effect:** $-OH$, $-OR$, $-NH_2$, $-NHR$ donate electrons into ring via lone pairs.
- **-M effect:** $-COOH$, $-CHO$, $-NO_2$, $-CN$ withdraw electrons from ring.

■ **MNEMONIC:** MNEMONIC for +M groups (Ortho-Para directors): 'OH NO, NH₂ helps ring!' — they activate the ring.

1.5 Reaction Intermediates

Intermediate	Description	Hybridization	Stability Order
Carbocation (R^+)	3-bonded C, +ve charge	sp^2 hybridized	$3^\circ > 2^\circ > 1^\circ > CH_3^+$
Carbanion (R^-)	3-bonded C, -ve charge	sp^3 hybridized	$CH_3^- > 1^\circ > 2^\circ > 3^\circ$

Free Radical ($R\cdot$)	3-bonded C, unpaired e^-	sp^2 hybridized	$3^\circ > 2^\circ > 1^\circ > CH_3\cdot$
Carbene ($:C\cdot\cdot$)	2-bonded C, 2 non-bonding e^-	sp^2 or sp^3	Very reactive

■ **FORMULA/RULE: Carbocation stability: $3^\circ > 2^\circ > 1^\circ$ — More alkyl groups = more hyperconjugation & inductive stabilization.**

1.6 Types of Reactions

Mechanism Types	More Reaction Types
• Substitution (S): One group replaces another	• Addition (A): Two molecules join → one product
• SN1: Two steps, 3° , racemization	• Elimination (E): Atoms removed → double bond forms
• SN2: One step, 1° , inversion (Walden inversion)	• Rearrangement: Skeleton of molecule changes
• E1: Two steps, needs weak base	• Oxidation: Gain of O / loss of H
• E2: One step, needs strong base	• Reduction: Gain of H / loss of O

✓ **TIP: SN2 always inverts configuration (like an umbrella flipping). SN1 gives racemic mixture (50:50).**

Chapter 2

Hydrocarbons

Alkanes • Alkenes • Alkynes • Aromatic Compounds

2.1 Alkanes — The Simplest Hydrocarbons

General Formula: $C_n H_{2n+2}$ | Hybridization: sp^3 | All single bonds

Key Reactions of Alkanes:

Reaction	Equation	Note
Combustion	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	Oxidation; energy released
Halogenation	$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (UV light)	Free radical substitution (FRS)
Pyrolysis	Higher alkane $\rightarrow$ smaller alkane + alkene	Breaking C–C bond by heat
Isomerization	n -butane $\rightarrow$ isobutane (catalyst, heat)	Structural change

■ **MNEMONIC: Free Radical Substitution steps: Initiation $\rightarrow$ Propagation $\rightarrow$ Termination (IPT). Initiator is UV light or heat.**

2.2 Alkenes — Double Bond Hydrocarbons

General Formula: $C_n H_{2n}$ | Hybridization: sp^2 | One C=C double bond

Reaction	Equation/Product	Key Point
Hydrogenation	$CH_2=CH_2 + H_2 \rightarrow CH_3-CH_3$ (Ni, heat)	Addition of H_2 (catalytic)
Halogenation	$CH_2=CH_2 + Br_2 \rightarrow CH_2Br-CH_2Br$	Decolourises brown Br_2 water
Hydrohalogenation	$CH_2=CH_2 + HX \rightarrow CH_3-CH_2X$	Markovnikov's Rule applies
Hydration	$CH_2=CH_2 + H_2O \rightarrow CH_3CH_2OH$ (H^+/SO_4^{2-})	Addition of water
Ozonolysis	Alkene + $O_3 \rightarrow$ aldehydes/ketones	Breaks C=C bond
Oxidation ($KMnO_4$)	Cold, dil. $KMnO_4 \rightarrow$ vicinal diol	Baeyer's test (pink $\rightarrow$ colourless)
Polymerization	$nCH_2=CH_2 \rightarrow -(CH_2-CH_2)_n-$ (polyethylene)	Addition polymerization

■ **FORMULA/RULE: MARKOVNIKOV'S RULE: In HX addition, H goes to the carbon that already has MORE hydrogens (i.e., the richer gets richer).**

✓ **TIP: Br_2/CCl_4 test: Alkene decolourises bromine water $\rightarrow$ confirms C=C double bond. Baeyer's test: pink $KMnO_4$ turns colourless.**

2.3 Alkynes — Triple Bond Hydrocarbons

General Formula: $C_n H_{2n-2}$ | Hybridization: sp | One $C \equiv C$ triple bond

Reaction	Equation	Note
Hydrogenation	$CH \equiv CH + 2H_2 \rightarrow CH_3CH_3$ (Pd/C, full)	Or stop at alkene using Lindlar's catalyst
Hydrohalogenation	$CH \equiv CH + HCl \rightarrow CH_2=CHCl$ (Markovnikov)	Vinyl chloride formed
Hydration	$CH \equiv CH + H_2O \rightarrow CH_3CHO$ (H_2SO_4 , $HgSO_4$)	Tautomerism involved
Acidic nature	$HC \equiv CH + NaNH_2 \rightarrow HC \equiv CNa + NH_3$	Terminal alkynes are acidic

■ FORMULA/RULE: Lindlar's catalyst (Pd + $CaCO_3$ + quinoline) stops at cis-alkene (partial hydrogenation). Na/liquid NH_3 gives trans-alkene.

2.4 Aromatic Compounds — Benzene & Friends

Benzene (C_6H_6) has 6 carbon atoms in a ring with alternating double bonds. It is stabilized by resonance (delocalization of π electrons).

Hückel's Rule: Aromatic compound must have $(4n+2)$ π electrons ($n = 0,1,2...$)

Reaction	Equation	Type
Electrophilic Sub. (EAS)	$\text{Benzene} + Br_2 \rightarrow \text{Bromobenzene} + HBr$ ($FeBr_3$ catalyst)	Halogenation
Nitration	$\text{Benzene} + HNO_3 \rightarrow \text{Nitrobenzene}$ (H_2SO_4 , 50–60°C)	Friedel-Crafts type
Sulfonation	$\text{Benzene} + H_2SO_4(\text{fuming}) \rightarrow \text{Benzene sulfonic acid}$	Reversible
Friedel-Crafts Alkylation	$\text{Benzene} + RCl \rightarrow \text{Alkylbenzene} + HCl$ ($AlCl_3$)	Carbocation attacks ring
Friedel-Crafts Acylation	$\text{Benzene} + RCOCl \rightarrow \text{ArCOR} + HCl$ ($AlCl_3$)	Acylium ion attacks ring

■ FORMULA/RULE: Activating groups (ortho-para directors): $-OH$, $-NH_2$, $-OCH_3$, $-CH_3$ (they are +M groups). Deactivating groups (meta directors): $-NO_2$, $-COOH$, $-SO_3H$, $-CHO$ (they are $-M$ groups). EXCEPTION: Halogens are deactivating BUT ortho-para directors.

■ MNEMONIC: MNEMONIC: 'OH NH OR CH activate para/ortho' — these are your friends (ring activators)!

Chapter 3

Haloalkanes and Haloarenes

R-X and Ar-X Compounds

3.1 Haloalkanes (Alkyl Halides)

General Formula: R-X (X = F, Cl, Br, I)

Named by replacing '-e' of alkane with the halide prefix. E.g., CH₃Cl = Chloromethane.

Preparation:

- From alkanes + X₂ (UV light) → Free radical halogenation
- From alkenes + HX (Markovnikov) → Alkyl halide
- From alcohols + SOCl₂ / PCl₅ / PCl₃ → Best method (no rearrangement with SOCl₂)

Chemical Reactions of Haloalkanes:

Reaction	Equation	Key Point
SN2 with NaOH (aq)	$R-X + NaOH(aq) \rightarrow R-OH + NaX$	Nucleophilic Sub; 1° preferred
SN2 with KCN	$R-X + KCN \rightarrow R-CN + KX$	C-C bond formed (nitrile)
SN2 with AgCN	$R-X + AgCN \rightarrow R-NC$ (isocyanide)	Isocyanide formed
SN2 with KNO ₂	$R-X + KNO_2 \rightarrow R-ONO$ (nitrite ester)	O-attack
SN2 with AgNO ₃	$R-X + AgNO_3 \rightarrow R-NO_2$ (nitroalkane)	N-attack
E2 with KOH/alc	$R-X + KOH(alc) \rightarrow \text{Alkene} + HX$	Elimination; hot, conc. KOH
Grignard Reagent	$R-X + Mg \text{ (dry ether)} \rightarrow RMgX$	Very important!
Wurtz Reaction	$2R-X + 2Na \rightarrow R-R + 2NaX$	C-C bond forming

■ **FORMULA/RULE:** KCN → nitrile (R-CN); AgCN → isocyanide (R-NC). KNO₂ → nitrite ester; AgNO₃ → nitroalkane. **RULE:** K-salts give the ambident nucleophile attack from C/O; Ag-salts give opposite.

✓ **TIP:** Grignard Reagent (RMgX) reacts with: Aldehydes → 2° alcohol; Ketones → 3° alcohol; Formaldehyde (HCHO) → 1° alcohol; CO₂ → Carboxylic acid.

3.2 Haloarenes (Aryl Halides)

Halogen directly attached to benzene ring. C-X bond has partial double-bond character (lone pair of X delocalizes into ring) → shorter & stronger than alkyl halides.

- **Less reactive** than haloalkanes towards nucleophilic substitution.
- EAS: Halogens are ortho-para directors but ring deactivators.
- **Nucleophilic Aromatic Substitution:** Possible only when strong electron-withdrawing groups (–NO₂) are present ortho/para to the halogen.

■ **CAUTION: KEY DIFFERENCE:** Haloalkane C–X bond is easily broken (S_N2 possible). Haloarene C–X has resonance → bond is stronger → NO S_N2. Reactions need harsh conditions.

3.3 Uses & Environmental Impact

- Chloroform (CHCl₃): Anesthetic, solvent
- Iodoform (CHI₃): Antiseptic
- DDT: Pesticide (now banned; non-biodegradable, accumulates in food chain)
- Freons (CFCs): Refrigerants — deplete ozone layer, now replaced

Chapter 4

Alcohols, Phenols and Ethers

–OH and –O– Functional Groups

4.1 Alcohols — R–OH

Types: 1° (primary), 2° (secondary), 3° (tertiary) based on C attached to –OH

Reaction	Equation/Product	Note
With Na metal	$2R-OH + 2Na \rightarrow 2R-ONa + H_2\uparrow$	Acidic nature; Na displaces H
With PCl ₅	$R-OH + PCl_5 \rightarrow R-Cl + POCl_3 + HCl$	Test for –OH group
Dehydration (conc. H ₂ SO ₄)	$R-OH \rightarrow \text{alkene (170°C) or ether (140°C)}$	Temperature determines product
Oxidation (1°)	$1^\circ-OH \rightarrow \text{Aldehyde} \rightarrow \text{Carboxylic acid}$	Using K ₂ Cr ₂ O ₇ /H ₂ SO ₄
Oxidation (2°)	$2^\circ-OH \rightarrow \text{Ketone}$	Using K ₂ Cr ₂ O ₇ /H ₂ SO ₄
Oxidation (3°)	$3^\circ-OH \rightarrow \text{No oxidation easily}$	Tertiary resists oxidation
Lucas Test	$3^\circ-OH \text{ reacts instantly; } 2^\circ-OH \text{ slowly; } 1^\circ-OH: \text{ no reaction}$	ZnCl ₂ /conc. HCl
Esterification	$R-OH + R'COOH \xrightarrow{H^+} R'COOR + H_2O$	Acid catalyst; reversible

■ FORMULA/RULE: LUCAS TEST: 3°>2°>1° reactivity. Cloudiness (alkyl chloride) appears immediately (3°), within 5 min (2°), or NOT at room temp (1°).

4.2 Phenols — Ar–OH

Phenol (C₆H₅OH) is more acidic than alcohols but less acidic than carboxylic acids. The –OH lone pair delocalizes into the ring, weakening the O–H bond.

Reaction	Equation/Product	Note
With FeCl ₃	$\text{Phenol} + FeCl_3 \rightarrow \text{Violet/Purple color}$	Test for phenol
Kolbe's Reaction	$\text{Phenol} + CO_2 \xrightarrow{NaOH, 130^\circ C, \text{ pressure}} \text{Salicylate} \rightarrow \text{Salicylic acid}$	Carboxylation
Reimer-Tiemann	$\text{Phenol} + CHCl_3 + NaOH \rightarrow \text{Salicylaldehyde}$	Formylation (–CHO at ortho)
With Br ₂ /water	$\text{Phenol} \rightarrow \text{2,4,6-tribromophenol (white ppt)}$	Ring very reactive, no catalyst needed
Nitration (dil. HNO ₃)	$\text{Phenol} \rightarrow \text{o- and p-nitrophenol}$	Mild conditions, mixed product

■ FORMULA/RULE: Acidity order: Carboxylic acid > Phenol > Water > Alcohol > Terminal alkyne

4.3 Ethers — R–O–R'

Ethers are relatively inert. Diethyl ether ($\text{C}_2\text{H}_5\text{--O--C}_2\text{H}_5$) is common solvent.

- **Cleavage with HI (excess):** $\text{R--O--R}' \rightarrow 2\text{RI}$ (iodoalkanes formed)
- **Cleavage with HI (limited):** $\text{R--O--R}' \rightarrow \text{ROH} + \text{RI}$ (if $\text{R} \neq \text{R}'$)
- **Peroxide formation:** Ethers + $\text{O}_2 \rightarrow$ explosive peroxides (store carefully!)

■ FORMULA/RULE: Williamson Synthesis: Best method to make ethers. $\text{R--ONa} + \text{R}'\text{--X} \rightarrow \text{R--O--R}' + \text{NaX}$ ($\text{S}_{\text{N}}2$)

Chapter 5

Aldehydes, Ketones and Carboxylic Acids

Carbonyl Compounds — C=O

5.1 Aldehydes (–CHO) vs Ketones (C=O)

Carbonyl group: C=O. Aldehydes have H on carbonyl C; ketones have two C groups.

Reaction	Equation/Product	Note
Nucleophilic Addition (HCN)	$RCHO + HCN \rightarrow RCH(OH)CN$ (cyanohydrin)	Both aldehydes & ketones
With NaHSO ₃	$RCHO + NaHSO_3 \rightarrow$ white crystalline addition product	Test; ketones only if methyl ketone
With NH ₂ OH	$RCHO + H_2NOH \rightarrow$ Oxime ($RCH=N-OH$) + H ₂ O	Condensation reaction
With N ₂ H ₄	$RCHO + H_2N-NH_2 \rightarrow$ Hydrazone + H ₂ O	Condensation
With 2,4-DNP	$RCHO + 2,4-DNP \rightarrow$ orange/yellow ppt	Test for C=O
With Grignard	$RCHO + R'MgX \rightarrow$ 2° alcohol (after hydrolysis)	Nucleophilic addition
Aldol Condensation	$2CH_3CHO \rightarrow$ $CH_3CH(OH)CH_2CHO$ (dil. NaOH)	Aldehydes with α-H
Cannizzaro Reaction	$2HCHO + NaOH \rightarrow CH_3OH + HCOONa$	Aldehydes WITHOUT α-H

5.2 Tests to Distinguish Aldehydes from Ketones

Test	Observation	Works for	Aldehyde	Ketone
Tollens' Test	Silver mirror on test tube (Ag deposited)	Aldehydes only	PASS	FAIL
Fehling's Test	Brick-red precipitate (Cu ₂ O)	Aldehydes (aliphatic)	PASS	FAIL
Schiff's Test	Pink/magenta color with Schiff's reagent	Aldehydes only	PASS	FAIL
NaHSO ₃ Test	White crystalline ppt	Ald. + methyl ketones	PASS	PARTIAL
Iodoform Test	Yellow ppt CHI ₃	CH ₃ CO–, CH ₃ CHOH–	CH ₃ CHO	CH ₃ COCH ₃

■ **FORMULA/RULE: IODOFORM TEST:** Positive for: CH_3CHO (ethanal), CH_3COCH_3 (acetone), $\text{CH}_3\text{CH}_2\text{OH}$ (ethanol), secondary alcohols of type $\text{CH}_3\text{CHOH-R}$. Always gives yellow CHI_3 precipitate.

5.3 Carboxylic Acids ($-\text{COOH}$)

Acidic character: $-\text{COOH}$ donates H^+ . Electron-withdrawing groups increase acidity.

Reaction	Equation/Product	Note
Salt formation	$\text{RCOOH} + \text{NaOH} \rightarrow \text{RCOONa} + \text{H}_2\text{O}$	Strong base neutralizes acid
Esterification	$\text{RCOOH} + \text{R}'\text{OH} \xrightarrow[\text{(H}^+, \text{heat)}]{\text{}} \text{RCOOR}' + \text{H}_2\text{O}$	Reversible; Fischer esterification
With PCl_5	$\text{RCOOH} + \text{PCl}_5 \rightarrow \text{RCOCl} + \text{HCl} + \text{POCl}_3$	Makes acid chloride
Decarboxylation	$\text{RCOONa} + \text{NaOH} \xrightarrow[\text{(CaO, heat)}]{\text{}} \text{RH} + \text{Na}_2\text{CO}_3$	Soda lime reaction
Hell-Volhard-Zelinsky	$\text{RCOOH} + \text{Br}_2 \xrightarrow{\text{(P)}} \alpha\text{-bromo acid}$	α -halogenation
Reduction (LiAlH_4)	$\text{RCOOH} \rightarrow \text{RCH}_2\text{OH}$ (primary alcohol)	Strong reducing agent

5.4 Derivatives of Carboxylic Acids

Reactivity: Acid chloride > Anhydride > Ester > Amide (decreasing order)

- Acid chloride (RCOCl): Most reactive, made by $\text{RCOOH} + \text{PCl}_5$
- Acid anhydride ($(\text{RCO})_2\text{O}$): Made by $2\text{RCOOH} \rightarrow \text{anhydride} + \text{H}_2\text{O}$ (dehydration)
- Ester (RCOOR'): Fruity smell, made by esterification
- Amide (RCONH_2): Made from ester + NH_3 or acid chloride + NH_3

Chapter 6

Amines

Nitrogen-Containing Organic Bases

6.1 Structure & Classification

Amines = Derivatives of NH_3 where H atoms are replaced by alkyl/aryl groups.

Primary & Secondary	Tertiary & Aromatic
Primary (1°): R-NH_2 (one R group)	Secondary (2°): R_2NH (two R groups)
Aliphatic: Alkyl groups attached to N	Aromatic (aryl amine): Ar-NH_2 (aniline)
Lone pair on N makes amines BASIC	Tertiary (3°): R_3N (three R groups)
Amines are Lewis bases (donate e^-)	Quaternary ammonium: R_4N^+ (4 groups, no lone pair — NOT a base)

6.2 Basic Character of Amines

Basic strength depends on availability of lone pair on nitrogen. More electron density on N $\rightarrow$ stronger base.

Order of Basicity (Aliphatic in gas phase): $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$

Order of Basicity (in aqueous solution): $2^\circ > 1^\circ > 3^\circ > \text{NH}_3$ (solvation effect)

Aromatic amines (aniline) are WEAKER bases than aliphatic amines.

■ **FORMULA/RULE: WHY aniline is a weak base:** lone pair on N delocalizes into benzene ring $\rightarrow$ less available for accepting proton $\rightarrow$ weaker base. Electron-withdrawing groups on ring further decrease basicity.

6.3 Reactions of Amines

Reaction	Equation/Product	Note
With HCl	$\text{R-NH}_2 + \text{HCl} \rightarrow \text{R-NH}_3^+\text{Cl}^-$ (salt)	Basic character
With Acyl chloride	$\text{R-NH}_2 + \text{RCOCl} \rightarrow \text{RNHCOR}$ (amide) + HCl	Acylation reaction
With Benzenesulfonyl Cl	$1^\circ\text{-NH}_2 + \text{C}_6\text{H}_5\text{SO}_2\text{Cl} \rightarrow \text{NaOH}$ soluble sulfonamide	Hinsberg test
With $\text{CHCl}_3 + \text{KOH}$	Primary amine $\rightarrow$ isocyanide (R-NC) foul smell	Carbylamine reaction
Diazotization (1°Ar)	$\text{ArNH}_2 + \text{NaNO}_2 + \text{HCl} (0-5^\circ\text{C}) \rightarrow \text{ArN}_2^+\text{Cl}^-$	Diazonium salt
Coupling reaction	$\text{ArN}_2^+ + \text{phenol/aniline} \rightarrow \text{azo dye}$ (orange/red)	Used in dyes

■ **FORMULA/RULE: CARBYLAMINE REACTION:** Only PRIMARY amines give isocyanide (foul smell) with CHCl_3/KOH . This is a TEST for primary amines.

✓ TIP: HINSBERG TEST: 1° amine → soluble sulfonamide in NaOH. 2° amine → insoluble. 3° amine → no reaction.

6.4 Preparation of Amines

- **Gabriel phthalimide synthesis:** Used to make pure 1° amines; phthalimide + RX + KOH + H₂O → R-NH₂
- **Hofmann bromamide reaction:** RCONH₂ + Br₂ + NaOH → R-NH₂ (1 less C than amide)
- **Reduction of nitro compounds:** ArNO₂ + 6H (Fe/HCl or Sn/HCl) → ArNH₂
- **Reduction of nitriles:** RCN + 4H → RCH₂NH₂ (using LiAlH₄)

Chapter 7

Organic Compounds Containing Nitrogen

Nitro compounds, Diazonium salts, Cyanides, Isocyanides

7.1 Nitro Compounds (R-NO₂)

Structure: N is central, bonded to O atoms and organic group. -NO₂ is a strong -M, -I group.

Reaction	Equation/Product	Note
Reduction (acidic)	$\text{ArNO}_2 + 6\text{H} (\text{Fe}/\text{HCl}) \rightarrow \text{ArNH}_2$	Primary aromatic amine formed
Reduction (neutral)	$\text{ArNO}_2 + 4\text{H} (\text{Zn}/\text{NH}_4\text{Cl}) \rightarrow \text{Ar-NHOH}$ (phenylhydroxylamine)	Partial reduction
Reduction (alkaline)	$\text{ArNO}_2 + 2\text{H} (\text{Zn}/\text{NaOH}) \rightarrow \text{Ar-N=N-Ar}$	Azobenzene formed
Halogenation	Nitrobenzene $\rightarrow$ meta-product preferentially	-NO ₂ is meta-director

7.2 Diazonium Salts (Ar-N₂⁺X⁻)

Diazonium salts are prepared at 0–5°C (ice-cold temperature is CRUCIAL — they decompose above 5°C). They are highly reactive and useful intermediates.

Reaction	Equation	Note
$\rightarrow \text{ArF}$ (Balz-Schiemann)	$\text{ArN}_2^+\text{BF}_4^- \rightarrow \text{ArF} + \text{N}_2 + \text{BF}_3$	Fluorobenzene preparation
$\rightarrow \text{ArCl}$ (Sandmeyer)	$\text{ArN}_2^+\text{Cl}^- + \text{CuCl} \rightarrow \text{ArCl} + \text{N}_2$	Cu-salt catalyst
$\rightarrow \text{ArBr}$ (Sandmeyer)	$\text{ArN}_2^+\text{Br}^- + \text{CuBr} \rightarrow \text{ArBr} + \text{N}_2$	Cu-salt catalyst
$\rightarrow \text{ArI}$	$\text{ArN}_2^+ + \text{KI} \rightarrow \text{ArI} + \text{N}_2$	No catalyst needed
$\rightarrow \text{ArCN}$ (Sandmeyer)	$\text{ArN}_2^+ + \text{CuCN} \rightarrow \text{ArCN} + \text{N}_2$	Cyanobenzene (nitrile)
$\rightarrow \text{ArOH}$	$\text{ArN}_2^+ + \text{H}_2\text{O} (\text{warm}) \rightarrow \text{ArOH} + \text{N}_2 + \text{HX}$	Hydrolysis
$\rightarrow \text{ArH}$	$\text{ArN}_2^+ + \text{H}_3\text{PO}_2 \rightarrow \text{ArH} + \text{N}_2$	Deamination
Coupling $\rightarrow$ Azo dye	$\text{ArN}_2^+ + \text{Ar}'\text{OH} \rightarrow \text{Ar-N=N-Ar}'\text{OH}$ (orange)	pH controlled

■ **FORMULA/RULE: SANDMEYER REACTION** uses CuCl/CuBr/CuCN as catalyst. ArI and ArH use KI and H₃PO₂ respectively — NO copper needed.

■ **MNEMONIC: MNEMONIC for diazonium replacements:** 'Friendly Clowns Bring Ice Cream, Oh Happy Day' $\rightarrow$ F (Balz-Schiemann), Cl (Sandmeyer), Br (Sandmeyer), I (KI), CN (Sandmeyer), OH (hydrolysis), H (H₃PO₂)

7.3 Cyanides and Isocyanides

Cyanides	Isocyanides
Cyanides ($R-C\equiv N$ / Nitriles)	Isocyanides ($R-N\equiv C$ / Isonitriles/Carbylamines)
• Made: $R-X + KCN \rightarrow R-CN$ (SN2)	• Made: $R-X + AgCN \rightarrow R-NC$
• Hydrolysis: $R-CN + H_2O \rightarrow RCOOH$	• Foul, unbearable smell
• Reduction: $R-CN + H_2 \rightarrow RCH_2NH_2$	• Also from: 1° amine + $CHCl_3 + KOH$
• C–C bond formation (chain lengthening)	• Hydrolysis: $R-NC + H_2O \rightarrow RNHCHO$ (formamide)

7.4 Quick Comparison Table — Nitrogen Compounds

Compound	Formula	Key Reactions	Applications
Nitro compound	$R-NO_2$	Nucleophilic/electrophilic reactions	Explosives, dyes
Amine	$R-NH_2$	Basic, acylation, Hinsberg test	Drugs, dyes, nylon
Diazonium salt	$ArN_2^+X^-$	Sandmeyer, coupling	Azo dyes, synthesis
Nitrile	$R-CN$	Hydrolysis, reduction, Grignard	Synthesis intermediate
Isocyanide	$R-NC$	Carbylamine test (foul smell)	Test for 1° amines
Amide	$RC(=O)NH_2$	Hofmann degradation, hydrolysis	Nylon, proteins

GOC	- Hybridization: sp^3=tetrahedral, sp^2=planar, sp=linear • $+I > -I$; $+M$ donates, $-M$ withdraws into ring - Carbocation stability: $3^\circ > 2^\circ > 1^\circ$ • $SN1$ = racemization; $SN2$ = inversion (Walden)
Hydrocarbons	Alkane + $X_2/h\nu$ → Free radical substitution • Alkene + HX → Markovnikov product • Lindlar's catalyst → cis-alkene (partial hydrogenation of alkyne) • Benzene: EAS; ortho-para directors activate; meta directors deactivate • Halogens = deactivating BUT ortho-para directors (EXCEPTION!)
Haloalkanes	$SN2$ preferred for 1°; $SN1$ preferred for 3° • KCN → nitrile; $AgCN$ → isocyanide • KNO_2 → nitrite ester; $AgNO_2$ → nitroalkane • Grignard ($RMgX$): $HCHO$ → $1^\circ alc$; $RCHO$ → $2^\circ alc$; R_2CO → $3^\circ alc$; CO_2 → $RCOOH$
Alcohols/Phenols	- Lucas test: 3° instant, 2° slow, 1° no turbidity ($ZnCl_2/HCl$) • $FeCl_3$ gives violet color with phenols - Acidity: $RCOOH > Phenol > H_2O > ROH$ • Kolbe reaction: Phenol + CO_2 → Salicylic acid; Reimer-Tiemann → Salicylaldehyde
Carbonyl Compounds	Aldehydes give Tollens' (silver mirror), Fehling's (red ppt), Schiff's (pink) • Iodoform test: CH_3CHO, CH_3COCH_3, CH_3CH_2OH → yellow CHI_3 ppt • Aldol condensation: needs α-H; Cannizzaro: no α-H ($HCHO$, $PhCHO$) • Acid derivatives reactivity: Acid chloride > Anhydride > Ester > Amide
Amines	Carbylamine test: only 1° amines → foul isocyanide smell • Hinsberg test: 1° → soluble sulfonamide; 2° → insol; 3° → no reaction • Basicity: Aliphatic $2^\circ > 1^\circ > 3^\circ$ (aqueous); Aromatic amines weakest • Gabriel synthesis → pure 1° amines; Hofmann degrades amide → amine
Diazonium & N-Compounds	Diazotization: $0-5^\circ C$ only! $ArNH_2 + NaNO_2 + HCl \rightarrow ArN_2^+Cl^-$ • Sandmeyer: Cl, Br, CN (needs Cu); ArI (needs KI); ArH (needs H_3PO_2) • Azo coupling: diazonium + phenol/aniline → orange/red azo dye • Nitro compound + Fe/HCl → Primary amine

■ **MNEMONIC: YOU GOT THIS!** Read each chapter once slowly, then use this revision sheet daily. Organic chemistry is about PATTERNS — once you see the pattern, every reaction makes sense. Best of luck for your PNST exam! ■